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Nitte (Deemed to be University)

(Established under Sec. (3) of UGC ACT, 1956)

First Semester M. Tech (CBCS) Degree Examinations, December 2025

25EVT133 – DIGITAL CONTROLLER

(For EVT)

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit-I** and **Unit-II** and **One full** question from **Unit-III**.

	Unit – I	Marks	BT*	CO*	PO*
1. a)	Explain the architecture of the dsPIC33C family digital signal controller with a neat block diagram.	7	L2	1	2
b)	Apply the concept of modulo addressing to an FIR filter implementation explain its working with an example memory structure.	7	L3	1	2
c)	Compare the interrupt processing mechanism of dsPIC33C with that of a conventional ARM Cortex-M microcontroller.	6	L2	2	5
2. a)	Explain the memory hierarchy and address modes of dsPIC33C and justify how they support high-speed DSP execution.	7	L2	1	5
b)	Using a simple program flow, demonstrate how to configure and enable an interrupt for an external event in dsPIC33C.	7	L3	1	5
c)	Summarize the applications of the dsPIC family in modern control systems.	6	L2	1	2
3. a)	Build a simplified block diagram of the dsPIC33/PIC24 Enhanced CPU and explain how each component contributes to DSP operations.	7	L3	1	2
b)	Illustrate the organization of the Interrupt Vector Table (IVT) in dsPIC33C controllers.	7	L2	2	5
c)	Outline the function of interrupt control and status registers in managing interrupt operations.	6	L2	2	5
	Unit – II				
4. a)	Summarize the different types of ADCs based on their conversion techniques.	6	L2	3	2
b)	Identify the impact of power-saving modes on Timer 1 functionality.	7	L3	4	5
c)	Analyze the architecture of a High-Resolution PWM (HRPWM) module with fine edge placement.	7	L4	4	5
5. a)	Outline the function of PWM Generator registers and their role in controlling signal timing.	6	L2	4	5
b)	Develop a code sequence to configure Timer 1 in capture/compare mode.	7	L3	4	2
c)	Distinguish between software-based and hardware-based logic implementations using CLC.	7	L4	3	5
6. a)	Illustrate the conversion sequence followed by a 12-bit ADC.	6	L2	3	5
b)	Demonstrate how Timer 1 can be utilized for PWM generation.	7	L3	4	5



- c) Analyze the influence of lock and write restrictions on PWM configuration security.

7 L4 4 2

Unit – III

7. a) Interpret how the queue mechanism improves SPI communication efficiency.
 b) Utilize the memory map registers to configure the clock speed and data rate in I²C.
 c) Compare the frame structure and performance between Classical CAN and CAN FD.
8. a) Outline the key registers used in the CAN FD module configuration.
 b) Illustrate how to utilize the memory map of QSPI to handle queued data transfers.
 c) Distinguish I²C protocol from SPI and QSPI in terms of hardware complexity and performance.

6 L2 5 2

7 L3 5 5

7 L4 5 5

6 L2 5 2

7 L3 5 5

7 L4 5 5

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
